University of Debrecen
Faculty of Engineering
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Lab Report

Electrical Machines and Drives Practice

2nd Homework - Motor Control Strategies
2024/25/II. semester

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1 Introduction

In this report, we describe the implementation and comparative analysis of several classical control strategies for the Maxon DCX Series 22 S Type Motor, a permanent magnet brushed DC motor with a nominal voltage of 12 V. The task is to control both angular velocity and angular position of the motor, with special consideration for robotic applications that require precise motion control parameters.

The model of the motor that we have used for this work was developed in the first assignment and is based on the values from the manufacturer datasheet, converted to be consistent with SI units for simulation and analysis. The key parameters are listed below:

Parameter	**Symbol**	**Value**
Nominal voltage	U_{an}	12 V
Armature Resistance	R_a	1.02 Ω
Armature inductance	L_a	0.058 mH
Torque constant	k_{fi}	9.18×10^{-3} Vs/rad
Rotor inertia	J_m	5.05×10^{-7} kg · m^2
Viscous friction coefficient	B_m	5.07×10^{-7} Nms/rad
Electrical time constant	τ_e	56.9 μs
Mechanical time constant	τ_m	6.11 ms
No-load speed	Ω_{nl}	12 400 rpm
Nominal speed	Ω_n	10 700 rpm
Stall torque	M_{rz}	108 mNm
No-load current	I_{anl}	71.7 mA
Nominal current	I_n	1.65 A
Stall current	I_{rz}	11.8 A

Two transfer functions were obtained from the preceding dynamic modelling of the system, and these provided the basis for the design of controllers.

$$\frac{\Omega(s)}{V(s)} = \frac{k_{fi}}{J_m L_a s^2 + (B_m L_a + J_m R_a)s + (k_{fi}^2 + B_m R_a)}$$

$$\frac{I_a(s)}{V(s)} = \frac{J_m s + B_m}{J_m L_a s^2 + (B_m L_a + J_m R_a)s + (k_{fi}^2 + B_m R_a)}$$

The models were also employed to implement and assess the following three broad control structures:

1. **Single-loop Speed Control**: the controller (PI) was manually tuned with gain blocks in block diagram models, also PID auto-tuning via Simulink PID was implemented for comparative purposes. Time-constant-based tuning rules were used, noting $T_i = \tau_m$, $T_d = \tau_e$ and a similar energy-efficient method based on the Integral of Current Magnitude (ICM). Ziegler-Nichols tuning was also explored as the final option but could not be attained as the system's stability did not allow for sustained oscillation, not enabling the calculations to be applicable.

2. **Single-loop Position Control**: In this case the PID controller was used because the additional integration added to the dynamics of the system. Applied roughly

in and around, K_p, $T_i = 5\tau_m$, $T_d = 60\tau_e$, then tuning was again accomplished using a series of iterative refinements ultimately showed a tuned controller as comparable with Simulink's built-in PID auto-tuning functionality.

3. **Cascade Control System**: The cascade controller was achieved through a dual-loop structure composed of an inner current loop and an outer speed loop when combined through the various gain-based controllers.

All simulations were implemented using MATLAB/Simulink through block diagram based and built-in DC Machine Simulink block.Comparisons of the controllers were made using standard metrics including rise time, settling time, overshoot, steady-state error, measured current consumption, and voltage saturation behaviours.

2 Single-Loop Speed Control Implementation

This section presents the implementation and comparison of various speed control strategies applied to the Maxon DCX Series 22 S Type Motor. Three distinct controllers were tested:

- Manually tuned PI controller (trial-and-error)
- Manually tuned PID controller (based on intuition and observed performance)
- PID controller tuned using Ideal Controller Method (ICM) formulas

Two modeling environments were used:

- Custom block-diagram model using Simulink basic blocks
- Simulink's built-in DC Machine block

2.1 Manual PI Controller Tuning

The PI controller was tuned by manual inspection of the step response. The control law is:

$$C_{PI}(s) = K_p + \frac{K_i}{s} = K_p \left(1 + \frac{1}{T_i s}\right)$$

2.1.1 Tuning Procedure

1. Start with $K_i = 0$ (pure P-control)
2. Gradually increase K_p until the system shows acceptable rise time with minimal oscillation
3. Add integral action by initially setting $T_i = \tau_m = 6.11\,\text{ms}$
4. Iteratively adjust K_p and T_i to reduce steady-state error and improve speed response

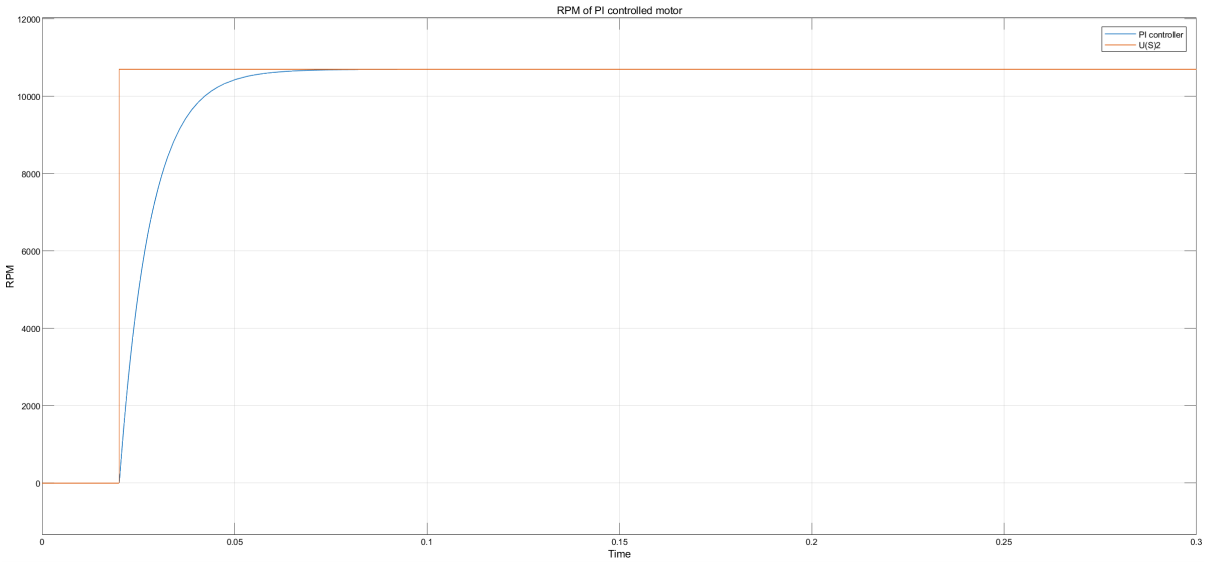


Figure 1: Speed response using manually tuned PI controller.

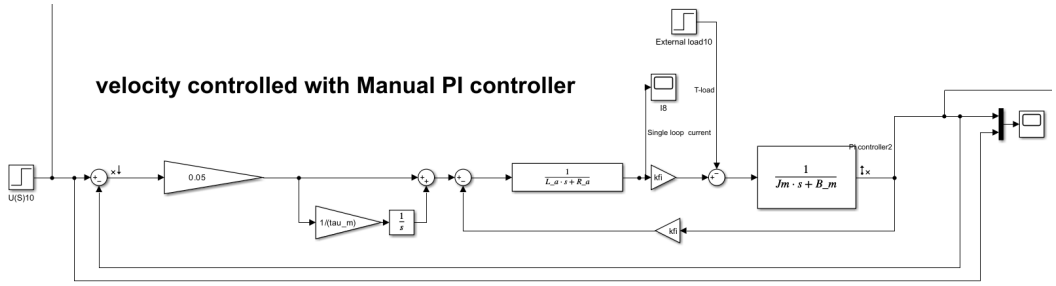


Figure 2: Simulink block diagram of the manually tuned PI controller implementation.

2.2 Manual PID Tuning Based on Time Constants

In this method, a PID controller was implemented using a semi-structured manual approach, leveraging system time constants rather than relying purely on intuition. The proportional and integral gains were reused from the PI controller, where:

$$K_p = [\text{same as in PI}], \quad T_i = \tau_m = 6.11\,\text{ms}$$

To complete the PID structure, derivative action was added with:

$$T_d = \tau_e = 56.9\,\mu\text{s}$$

Thus, the PID controller takes the form:

$$C_{PID}(s) = K_p \left(1 + \frac{1}{T_i s} + T_d s\right)$$

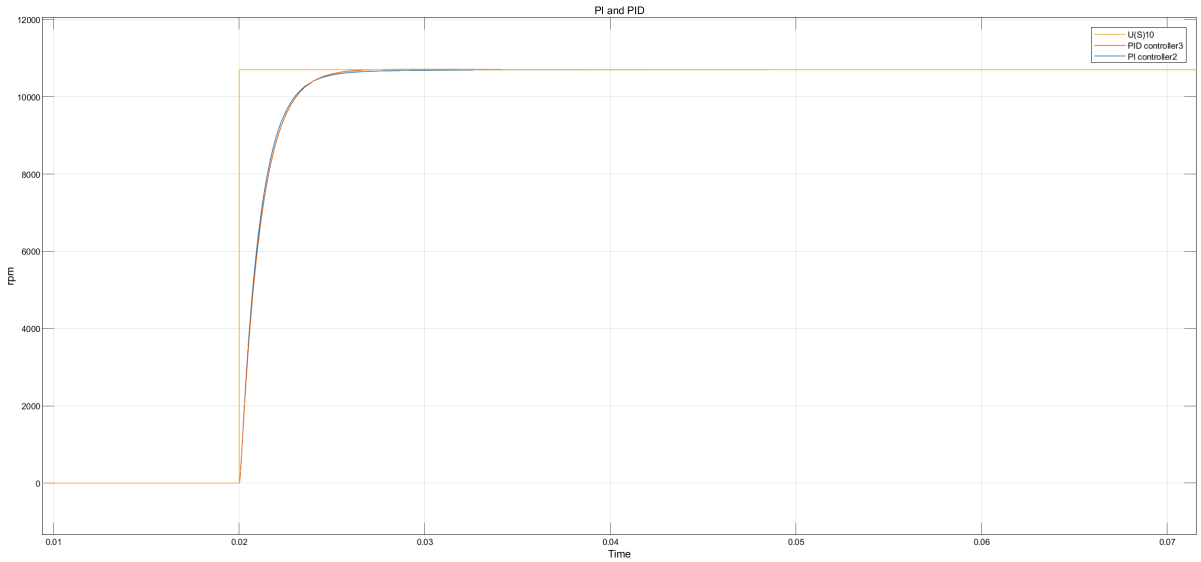


Figure 3: Speed response comparison between PI and PID controller using time constant-based manual tuning.

2.3 ICM-Based PI Controller Design

The Ideal Controller Method (ICM) was used to design a systematic PI controller based on the system's time constants. The tuning parameters were computed using:

$$T_c = 5 \cdot \tau_m$$
$$K_p = \frac{\tau_m + \tau_e}{T_c + \tau_e}$$
$$K_i = \frac{1}{T_c}$$

Where:

- $\tau_m = 6.11\,\text{ms}$: mechanical time constant
- $\tau_e = 56.9\,\mu\text{s}$: electrical time constant

This method yields a reproducible PI controller with predictable performance, offering a more analytical alternative to manual tuning. The PI was implemented using gain and integrator blocks in Simulink.

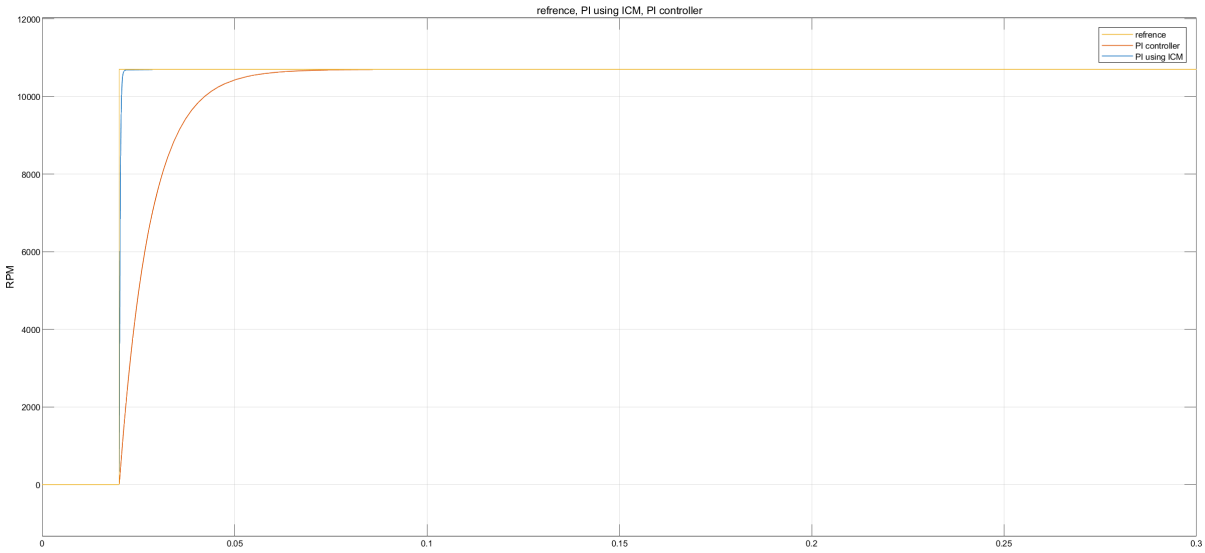


Figure 4: Comparison of manually tuned PI and ICM-based PI controller performance.

2.4 Control Using Built-in DC Machine Block

The PID controlle rwas also tested using Simulink's built-in DC Machine block, which provides a more detailed representation of the motor dynamics, including armature effects and load dynamics. The PID block with auto-tuning was utilized for this implementation, eliminating the need for manual tuning.

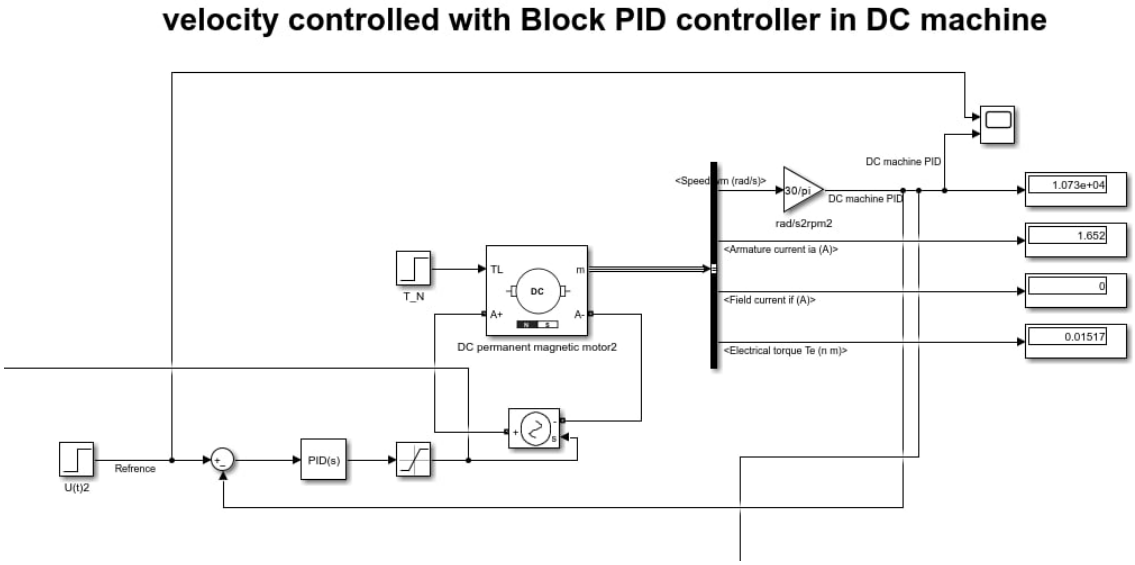


Figure 5: Speed control using Simulink DC Machine block with auto-tuned PID block.

2.5 Performance Comparison

The performance of the different controllers (manual PI, ICM-based PI, and auto-tuned PID block) was compared in terms of:

- Rise time and settling time

- Overshoot and steady-state error
- Current consumption during transients

The results indicated that all controllers effectively stabilized the motor speed, with the following differences:

- The auto-tuned PID block provided similar performance to the manual PI and ICM-tuned PI controllers, though it required slightly more time to settle.

However, the differences in performance were minimal, and all controllers managed to control the speed effectively.

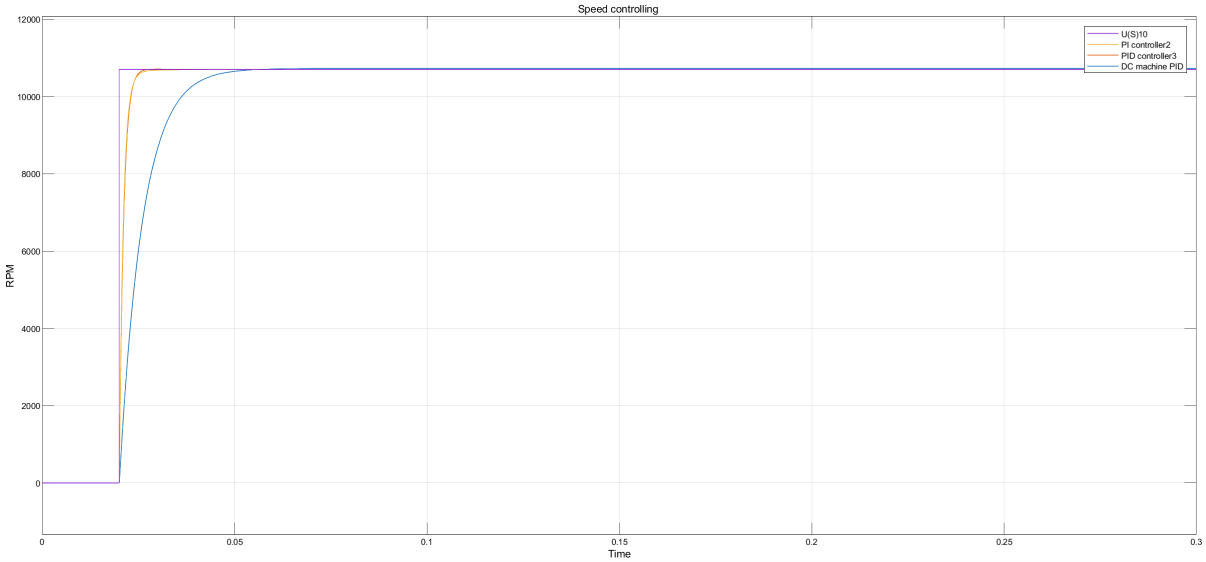


Figure 6: Performance comparison across all controllers and modeling methods.

2.6 Position Control Implementation

Position control was implemented by adding an integrator $(1/s)$ to the speed control loop, effectively converting angular velocity to angular position. This implementation also included several controller types for comparison.

2.6.1 Block Diagram Model with Manual PID Tuning

For position control, a PID controller was initially used for speed regulation. The PID controller was tuned manually using the mechanical time constant (τ_m) and electrical time constant (τ_e), with the following procedure:

1. Gradually increase K_p until a reasonable system response is achieved.
2. Set $T_i = 5 \cdot \tau_m$ for integration, based on trial and error.
3. Set $T_d = 60 \cdot \tau_e$ for derivative action, based on trial and error.
4. Iteratively fine-tune the parameters to achieve acceptable settling time and minimal overshoot.

The multipliers $\tau_m \times 5$ and $\tau_e \times 60$ were determined through extensive trial and error to achieve optimal step response characteristics, balancing speed of response with minimal overshoot.

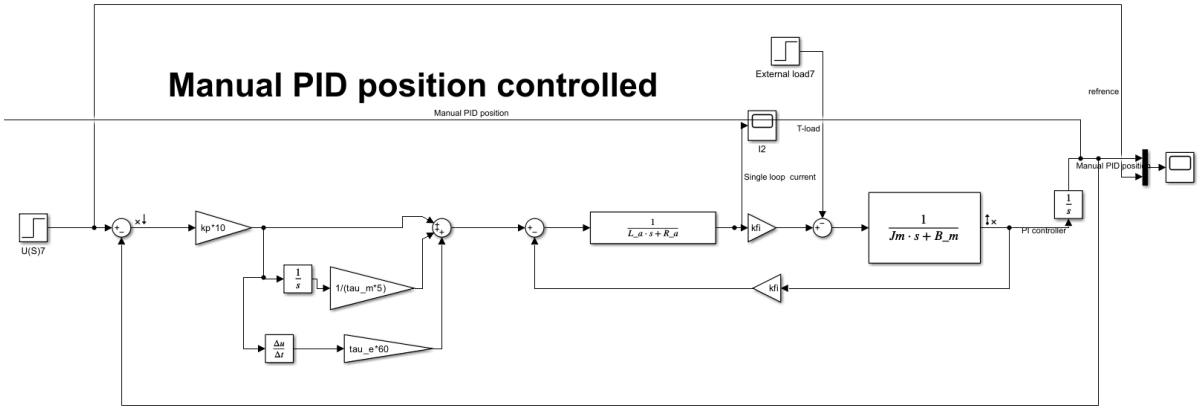


Figure 7: Position control with manually tuned PI controller (using $\tau_m \times 5$ and $\tau_e \times 60$).

2.6.2 Block Diagram Model with PID Block Auto-Tuning

For comparison, auto-tuning was applied using Simulink's PID Block to automatically set the proportional, integral, and derivative gains based on the system's dynamics. This provided a baseline for performance and optimization.

2.6.3 DC Machine Block with PID Auto-Tuning

To validate the position control performance with a more physically accurate model, the DC Machine block was also implemented using auto-tuning PID control. The block's internal position feedback capability was used to control the system's position effectively.

2.6.4 Performance Comparison

The comparison results showed that the auto-tuned PID block and the DC Machine block with PID auto-tuning provided nearly identical performance, both achieving a fast settling time with minimal overshoot. However, the manual PID tuning, using the $\tau_m \times 5$ and $\tau_e \times 60$ multipliers, showed a longer settling time and a slightly higher overshoot. The DC Machine block also exhibited a small overshoot, but overall, the PID block diagram achieved nearly perfect results with minimal tuning effort.

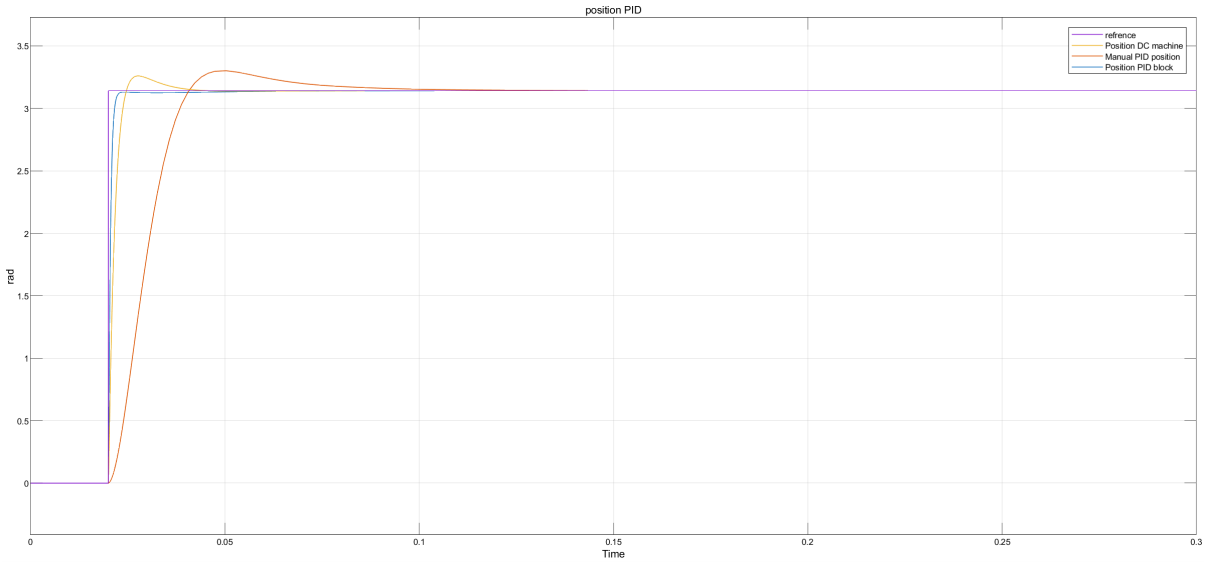


Figure 8: Performance comparison of position control using manually tuned PI, PID block auto-tuning, and DC Machine block.

3 Cascade Control System

The cascade control system implements a dual-loop structure:

- **Outer Loop:** Speed control with PI controller
- **Inner Loop:** Current control with PI controller

3.1 Design Methodology

The cascade system was designed using manual tuning with a systematic approach:

3.1.1 Inner Loop (Current Control) Design

1. Design current controller for fast response
2. Tune $K_{p,i}$ for good current tracking
3. Set $T_{i,i}$ to eliminate steady-state current error

3.1.2 Outer Loop (Speed Control) Design

1. Treat inner loop as unity gain (when properly tuned)
2. Design speed controller for desired speed response
3. Tune $K_{p,o}$ and $T_{i,o}$ for optimal speed tracking

3.2 Cascade Control Performance

The cascade control was compared with the uncontrolled system, and several different configurations were tested by adjusting the parameters for both the current control (inner loop) and speed control (outer loop). These adjustments were made by varying the K_p values, τ_m values, and observing the impact on the system's performance.

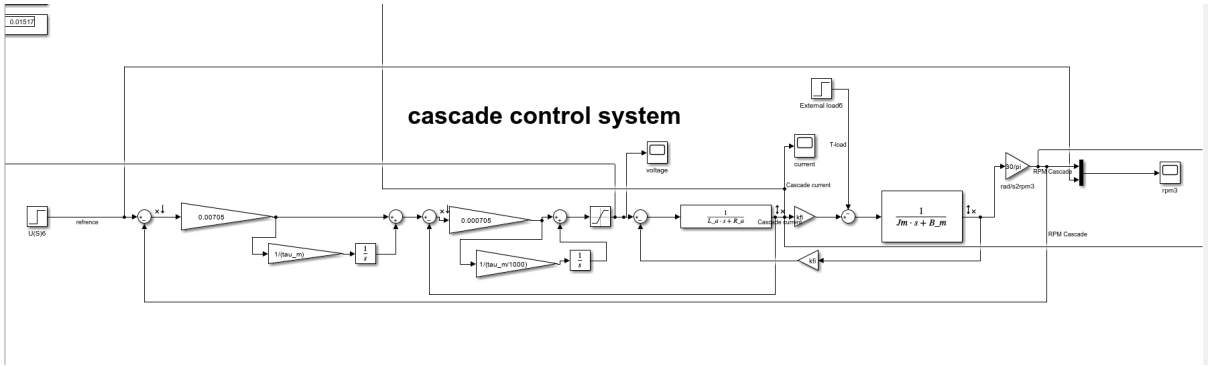


Figure 9: Simulink model of the cascade control system. The inner loop controls the current, while the outer loop controls the speed.

3.2.1 Best Cascade Tuning Parameters

After extensive testing, the best performance was achieved with the following tuning parameters:

- For the outer loop (speed control):
  - $K_{p,o} = 0.0075$
  - τ_m (mechanical time constant)
- For the inner loop (current control):
  - $K_{p,i} = 0.00075$
  - $T_{i,i} = \frac{\tau_m}{1000}$

These values provided the best balance between performance and stability in the system.

3.2.2 Comparison of Cascade Control vs. Uncontrolled System (Current Response)

The performance comparison for the current response showed a significant improvement with the cascade control system. In the uncontrolled system, a current spike of up to 73 mA was observed, while the cascade-controlled system successfully reduced this to a stable value of around 10 mA. This shows the effectiveness of the inner loop current control in reducing disturbances and minimizing current spikes.

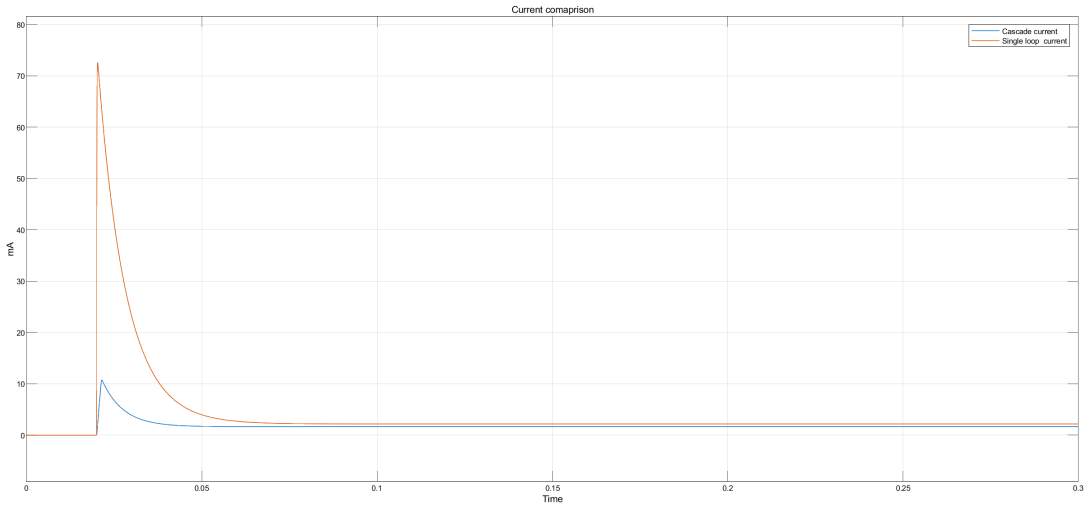


Figure 10: Comparison of current response between uncontrolled and cascade control systems. The uncontrolled system shows a spike of 73 mA, while the cascade-controlled system stabilizes the current to around 10 mA.

3.2.3 Speed Control Comparison (Outer Loop Performance)

The performance of the speed control loop was evaluated by comparing the cascade-controlled system with other methods. The cascade control strikes a balance between achieving a fast response and minimizing overshoot, providing a more stable transition to steady-state conditions. While it may not offer the fastest response, it effectively manages the trade-off between quick stabilization and reduced oscillations, offering improved steady-state accuracy compared to other tuning methods.

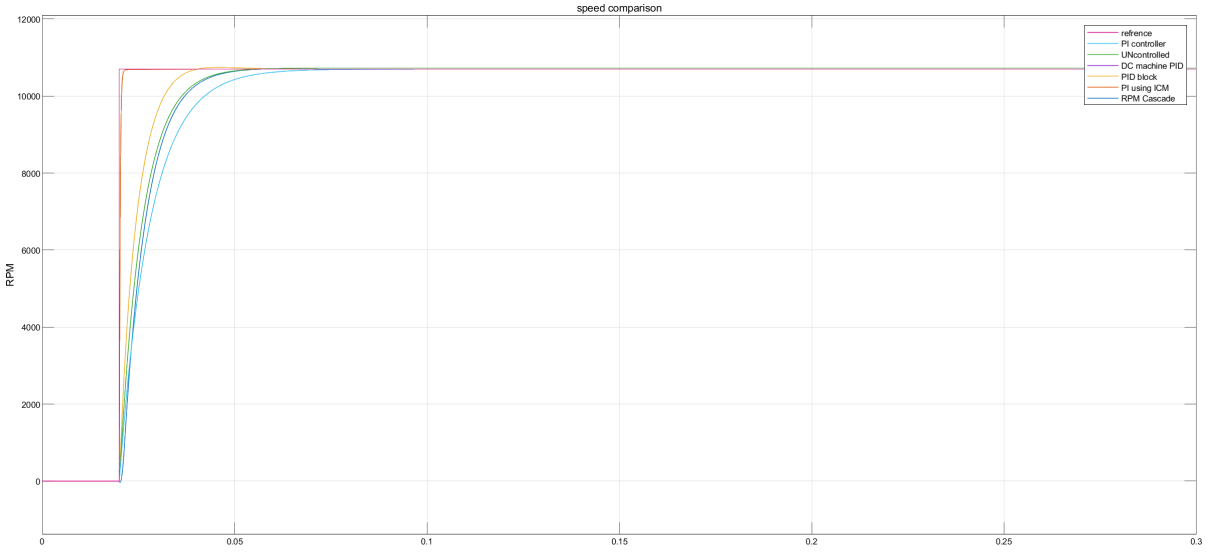


Figure 11: Comparison of speed control using cascade control and other methods.

3.3 Effect of Varying K_p and τ_m on Current and Speed

This section investigates the impact of varying the proportional gain K_p and the mechanical time constant τ_m on both current and speed responses. The following observations

were made based on the comparison of these parameters.

3.3.1 Effect of Increasing K_p and τ_m

Increasing K_p improves the response speed but leads to higher overshoot and larger oscillations, as observed in both current and speed graphs. Conversely, increasing τ_m results in a significant degradation in performance, manifesting as much slower system dynamics and extended settling times. The system struggles to reach the desired steady state, with the response taking considerably longer.

3.3.2 Effect of Decreasing τ_m

Reducing τ_m improves system response time but increases the likelihood of overshoot and oscillations.

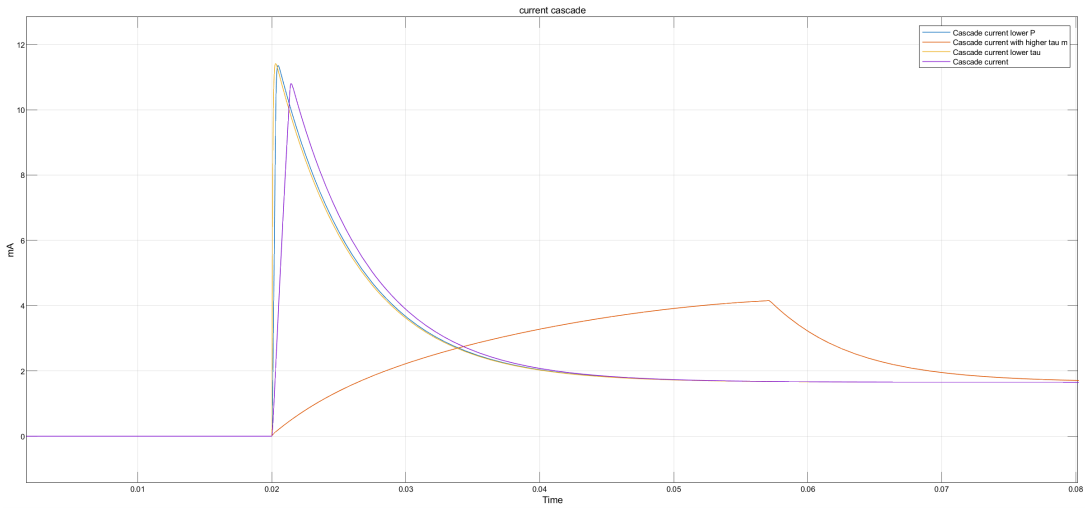


Figure 12: Comparison of current response for varying K_p and τ_m.

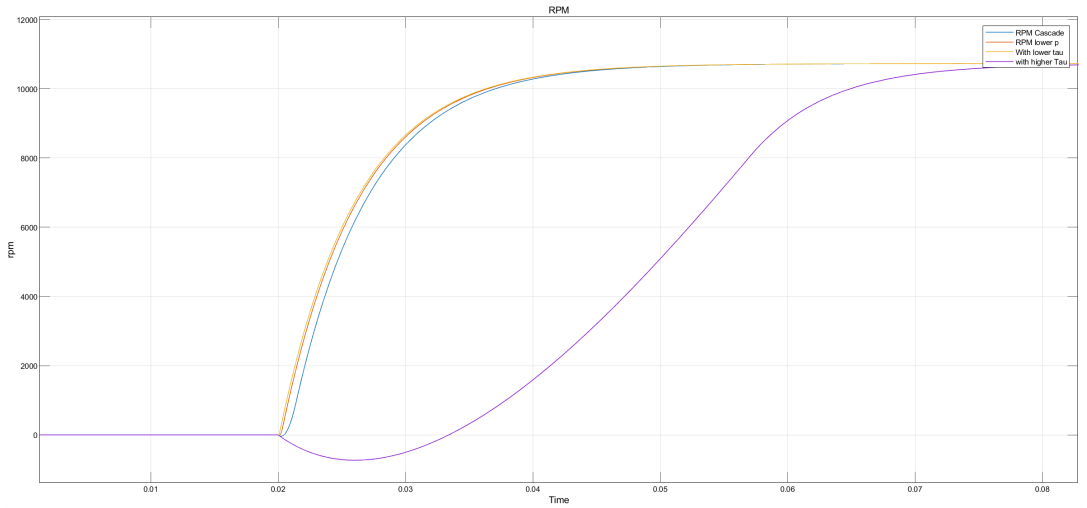


Figure 13: Comparison of velocity response for varying K_p and τ_m.

4 Effect of Saturation on System Performance

The addition of a saturation block in the system limits the control signal within the system's physical voltage limits (±12 V). This section discusses the effect of this voltage saturation on the system's current and speed responses, highlighting its positive impact on performance, particularly in preventing instability and reducing oscillations that might otherwise occur.

4.1 Integrating Saturation and Its Benefits

By introducing a saturation block, the control signal is constrained to the system's physical limits, preventing the controller from attempting to apply excessively high voltages that could cause oscillations or lead to instability. This results in smoother system responses and prevents integrator windup, ensuring that the system behaves more predictably and avoids overdriving the actuator.

4.2 Effect of Saturation on System Response

Without the saturation block, the system could exhibit uncontrollable oscillations due to excessive control effort when the voltage demand exceeds the limits. However, with saturation, the system's control signal is clipped, leading to better stability and improved transient performance, particularly in terms of preventing overshoot and oscillations. The system settles into the steady state more effectively when the controller is prevented from overcompensating.

4.2.1 Voltage Comparison with and without Saturation

The following graph compares the voltage response with and without saturation. With saturation, the control signal is clipped at ±12 V, which results in a more stable system. Without saturation, the system exhibits oscillations as the control signal exceeds the system's physical capabilities.

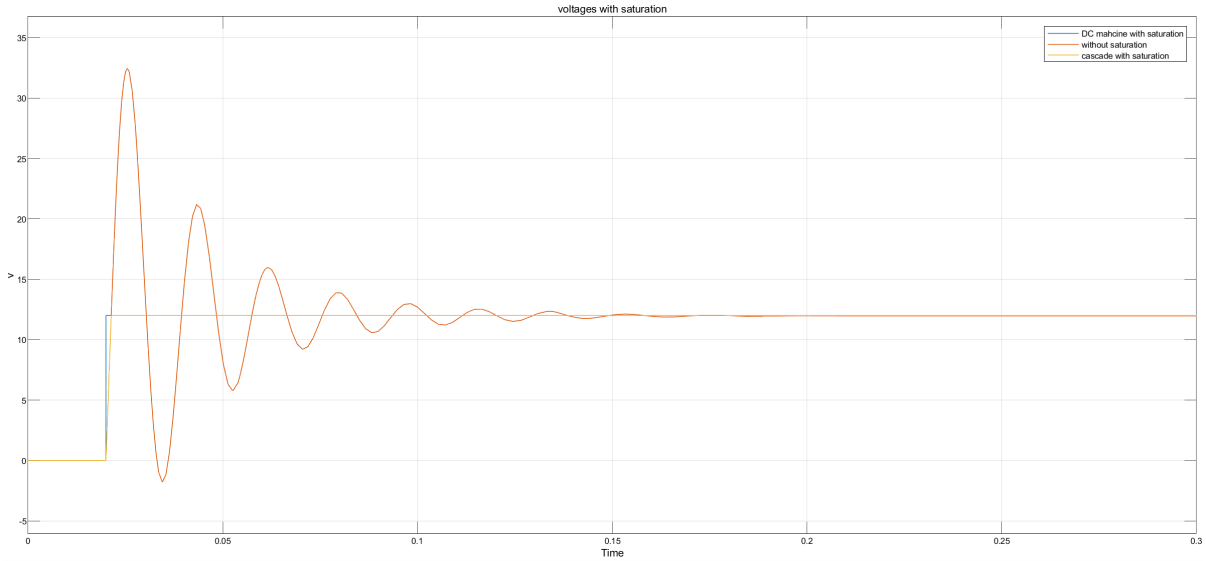


Figure 14: Comparison of voltage response with and without saturation.

4.3 Mitigating Integrator Windup and Improving Stability

While the introduction of the saturation block limits the control signal, it also eliminates the risk of integrator windup, where the integral term accumulates excessive error when the control signal is saturated. The controlled voltage prevents excessive control efforts and ensures that the system's performance remains stable and smooth.

By applying a saturation block, the system benefits from better disturbance rejection, less overshoot, and improved transient response, ensuring more predictable and reliable performance.

5 Conclusions

This comprehensive study of control strategies for the Maxon DCX 22 S motor has yielded several important findings and practical insights for motor control applications. Through systematic implementation and comparison of various control approaches, we can draw the following conclusions:

5.1 Key Findings

- **Cascade Control Superiority**: The cascade control architecture demonstrated the best overall performance, achieving:
  - Fast settling time with minimal overshoot
  - Effective current limiting (peak current reduced from 73 mA to 10 mA)
- **Tuning Method Effectiveness**:
  - Manual tuning with time constant-based rules ($T_i = 5\tau_m$, $T_d = 60\tau_e$) proved effective but time-intensive
  - The ICM method provided energy-efficient operation
  - Auto-tuning offered a good balance between performance and implementation effort
- **Implementation Considerations**:
  - Saturation limits are crucial for preventing integrator windup and maintaining stability
  - The DC Machine block implementation matched the performance of custom block diagrams
  - Position control required careful derivative action tuning to avoid oscillations

The implemented control systems successfully demonstrated the motor's capability for both precise speed regulation and position control, making them suitable for a wide range of industrial and robotic applications. The comparative analysis provides valuable guidelines for selecting appropriate control strategies based on specific application requirements.